

A User-Centric Exploration of Axiomatic Explainable AI in Participatory Budgeting.

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ABSTRACT

Explainable Artificial Intelligence (XAI) has been widely used to clarify the opaque nature of AI systems. One area where XAI has gained significant attention is Participatory Budgeting (PB). PB mechanisms aim to achieve a proper allocation concerning both the votes collected based on user's preferences and the budget. An essential criterion for evaluating these mechanisms is their ability to satisfy desired properties known as axioms. However, even though there are complex voting rules that meet some axioms, concerns regarding transparency persist. In this study, we propose an approach to provide explanations in a PB setting by treating axioms as constraints and seeking outcomes that adhere to these constraints. This method enhances system transparency and explainability. Each potential allocation is accepted or rejected based on whether it satisfies the axioms, and the linear nature of the axioms reduces computational complexity. We evaluated our approach with real-world users to assess its effectiveness and helpfulness. Our pilot study shows that users generally find explanations helpful for understanding the system's decisions and perceive the outcomes as fairer. Additionally, users prefer general explanations over counterfactual ones.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; • **Applied computing** → *Sociology*; • **Computing methodologies** → **Cooperation and coordination**.

KEYWORDS

Explainable Artificial Intelligence; Users; Social Choice; Participatory Budgeting.

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1 INTRODUCTION

The field of Explainable Artificial Intelligence (XAI) focuses on elucidating the reasoning behind decisions or predictions from Artificial Intelligence (AI) systems, thereby making AI models more understandable and transparent [14], [7]. Various motivations for using explanations in AI algorithms are cited in the literature [8], [6]. These reasons include: 1- Generating trust (trustworthy AI). 2- Meeting legal requirements. 3- Upholding social responsibility, fairness, and risk avoidance. 4- Creating accountable and reliable models. 5- Minimizing biases. 6- Validating models. To enhance transparency, several approaches have been proposed, such as feature importance, which identifies the most influential features in the model's decision-making process [10], [12], and counterfactual explanations, which involve generating alternative scenarios where the model's output would change [15], [13]. Another significant approach is the axiomatic approach, which uses axioms (a set of desired properties and agreed values) to justify the outcomes [9]. Our work focuses on the axiomatic approach, developing explainability based on this method for AI algorithms aimed at solving Participatory Budgeting (PB) problems.

PB is a democratic approach to determining the allocation of funds for public projects [2]. In this set of challenges, we encounter various projects with associated costs and a limited budget, typically insufficient to cover all projects. The goal is to decide which projects to fund and which to dismiss based on the votes of participants. In cities worldwide, such as Paris, Sydney, and New York, residents engage in participatory budgeting to collectively determine the allocation of public funds. Residents cast votes on proposals ranging from constructing a playground, repaving a road, and planting a community garden, to installing new street lights. The outcome is shaped by collective voting, determining the projects that secure funding and those that do not.

Several intricate methods exist to determine a PB outcome, considering both the budget constraints and the votes. Each of these methods adheres to a set of axioms. Generally, mechanisms that satisfy a higher number of axioms, representing desired properties, are deemed "better" and more justifiable. Therefore, axioms serve as criteria for guiding the allocation process and justifying the final outcome. Procaccia [11] notes that axioms capture what is desirable and should be applied to explain the mechanism's outcome to voters. Cailloux et al. [4] developed a logic-based language to describe the strengths and weaknesses of different single-winner

voting rules for voters. Boixel et al. [3] view finding an argument for an outcome as a constraint-solving problem in single-winner settings, where a triple of (variables, domain, constraints) needs definition.

In our work, for the first time, we designed a transparent mechanism based on axioms to calculate and explain the outcome of a PB setting, and evaluated the effectiveness of axiomatic explanation on real-world users. The proposed approach considers axioms as constraints to be fulfilled and systematically enumerates allocations that meet these constraints. Through this axiomatic approach, we can automatically provide reasoning and explanations for each chosen or excluded allocation based on the criteria of whether they satisfy axioms or not. In the following section, we will briefly discuss participatory budgeting modeling, the axioms employed, and formulate our approach.

2 PARTICIPATORY BUDGETING MODELING

Every PB settings can be modeled by the following variables, [1]:

- The first variable is agents (also known as residents or voters) that will be shown as $N = \{1, \dots, n\}$.
- The second variable is the budget (or resources). We show the budget as B , which is the money limitation we can spend on projects in general.
- There is also a set of projects (also known as candidates or alternatives) as $X = \{x_1, \dots, x_m\}$.
- The next consideration is the degree of completion, which can be divided into two general types: divisible and non-divisible. Divisible projects can be completed to different degrees (continuous). In our setting, all the projects are non-divisible. It means they only can be completed 100% (shown by 1) or 0 (shown by 0).
- Each project $x \in X$ has an associated cost function shown as: $cost : X \rightarrow \mathbb{R}^+$, and $cost(x_j)$ indicates the cost of the j -th project.
- Agents' votes demonstrate how agents can express their selection. In our setting, we use approval voting, which means each voter can approve or disapprove of a project (shown by 1 or 0, respectively). We consider that voters are indifferent between approved or disapproved projects. All the approved projects (shown by 1) are strictly preferred to not approved ones (shown by 0). We show our vote list as $Vote = \{vote^1, \dots, vote^n\}$. $vote^n$ shows n th voter ballot that contains m members of 1 or 0 indicating approval or disapproval of a project respectively.

Based on the previous notations, we can define a PB voting rule as follows:

$$F : \langle Vote, B, cost(x) \rangle \rightarrow \{C \mid C \subseteq 2^X, \sum_{c_k \in C} cost(c_k) \leq B\} \quad (1)$$

We show an allocation as C containing funded x_j s, and $cost(c_k)$ is a function that maps selected projects (c_k s) to their costs, and 2^X is all the possible allocations.

Axioms are agreed-upon values that we can use as the foundational principles for our voting mechanism and deploy them as explanations for users. In the following, we introduced the axioms that we used for the purpose of providing explanations:

- **Feasibility:** This axiom says that an outcome C is feasible if and only if: $\sum_{c_k \in C} cost(c_k) \leq B$. In other words, it says that the summation of our spending money should not be beyond our budget.
- **Exhaustiveness [2]:** This axiom says that a feasible allocation, C , is exhaustive if it is impossible to fund any more projects with the leftover budget [1]. Considering our setting characterization (approval voting and non-divisible degree of completion), we can formulate it as:
 $(B - \sum_{c_k \in C} cost(c_k)) < cost(c'), c' \in X \setminus C$
- **Utilitarian welfare_consistent Axiom:** An allocation $C \in 2^X$ is utilitarian welfare consistent if there exists no $C' \in 2^X$ such that $\sum_{i=1}^n u_i(C') > \sum_{i=1}^n u_i(C)$ [5]. This axiom states that an allocation that causes a bigger utility is preferred to other allocations.

3 METHODOLOGY

Leveraging axioms to determine outcomes offers the advantage of transparency. We employed axioms, which represent agreed-upon values, to advocate for funding certain projects while providing reasoning for the rejection of others.

The proposed mechanism operates based on a constraint-solving problem framework, which involves an objective function aimed at maximizing the number of funded projects, along with a set of linear constraints. We formulated each mentioned axiom as an integer linear form and used an Integer Linear Programming (ILP) solver to find all possible outcomes that meet our constraints.

In this setting, we present the outcome of funding the projects as X . If $x_j = 1$ means that the project j has been funded and if $x_j = 0$ means that this project has not been funded. Based on that, the problem can be formulated in ILP as follows:

$$\sum_{j=1}^m x_j \quad (2)$$

subject to:

$$x_j \in [0, 1] \quad \text{for all } j = 0, 1, \dots, m,$$

We applied our ILP problem in Python, Gurobi optimizer¹. Gurobi Optimizer is a prescriptive analytics platform and a decision-making technology.

4 EXPERIMENTAL USER STUDY

As the next step of our work, we tested the proposed axiomatic approach on real-world users to see how these explanations work in practice. More specifically, we studied whether presenting reasoning to users will bring more transparency from the user perspective and help them to better trust the system (trustworthiness).

For instance, in a simple scenario, the proposed approach might provide an explanation like this: "Although you supported project A, more voters voted for project B. Consequently, funding project B would result in greater overall satisfaction among voters. Additionally, our budget constraints prevent us from funding both projects." In this example, the proposed axiomatic approach employs the utilitarian welfare-consistent axiom (preferring allocations that maximize voter utility) and feasibility (ensuring the total project

¹<https://www.gurobi.com/>

costs stay within budget) to identify the most suitable allocation while generating reasoning for alternative scenarios.

Our user study primarily focuses on assessing whether explanations generated based on axioms enhance users' comprehension of the outcomes in the PB setting. Our arguments regarding the outcome derive from theoretical mathematical properties, and it remains uncertain whether users perceive them as understandable. For instance, while it is theoretically preferable to maximize the utility function, it is unclear from the user's perspective whether this argument is **understandable**. The secondary objective of this study is to examine how our explanations and justifications assist participants in:

- Better understanding the decision-making process of the system,
- Developing greater trust in the algorithm and the system's decisions,
- Perceiving the system's outcomes as fairer after receiving the provided explanations

During this assessment, we examined whether users lean toward personalized explanations or counterfactual explanations. On the one hand, a personalized explanation involves providing reasoning for why a particular outcome was chosen, offering insight into why certain projects were funded or not based on the user's votes. On the other hand, a counterfactual explanation steps into the realm of "what if," clarifying how the outcome would change if other participants' votes shifted to align with the user's preferences. It's important to note that these two types of explanations operate independently of each other.

Our hypotheses regarding the mentioned questions are listed below:

- (1) We hypothesize that as explanations will help users to better **understand** the decision of the system.
- (2) We hypothesize that our explanations will pave the way and help users to better **trust** the system.
- (3) We hypothesize that our explanations will help users perceive the outcome of the system as **fairer** compared to situations where we do not provide any explanation.
- (4) We anticipate that users prefer **personalized** explanations over **counterfactual** explanations.

Our user study began with the acquisition of human ethics research approval², an essential step in our study protocol. Following this, we disseminated information about our research through Mechanical Turk.

During the data collection phase, we engaged participants by providing an overview of the study and obtaining their consent to participate. Subsequently, participants were presented with our participatory budgeting problem and asked to vote based on their personal interests. Based on the collected votes, the algorithm's decision was revealed to them.

We then gauged participants' perceptions of the algorithm's clarity and trustworthiness through a series of questions. Participants were also provided with automated explanations and justifications, followed by inquiries regarding the transparency of the system post-explanation.

²HC reference number: iRECS5941

Upon completion of the survey, we proceeded to the phase of data validation and analysis, culminating in the publication of our results.

4.1 Survey Structure:

We designed a questionnaire to evaluate our hypotheses regarding explainable PB setting. The questionnaire is structured to maintain anonymity, gathering only essential demographic information such as gender, age, participants' country, and their familiarity with computers. This data collection serves to explore potential correlations between gender, the stage of development of the country, the level of computer science knowledge, and how participants assess explanations. Our study involved two distinct participatory budgeting cases, each with different scenarios. Each participant randomly assigned to one case and based on their vote, explanations regarding one scenario presented to them. We only showed one participatory problem to users in order to prevent potential experimental confusion and bias. The two participatory budgeting cases are independent of each other. Randomly, we presented one case to each user in a way that each case is shown 50% of the time.

In the following, we presented each case and possible options for the user to vote.

4.2 Presented Participatory Budgeting Cases:

• CASE 1:

Consider there is a budget of \$11,000 allocated to the Department of Computer Science, and the dean aims to allocate these funds to two potential projects:

- (1) Buying new books for the department library with a total cost of \$10,000.
- (2) Upgrading the computer labs, incurring a cost of \$9000. Given the available budget, the dean can only fund one out of these two projects. To facilitate this decision-making process, the dean has appointed three representatives for the department, and your role is to serve as the third representative, providing your vote.

We considered that the first and second representatives are Alice and Bob. Alice and Bob both approved only acquiring new books for the library. We are employing the greedy rule as our voting method. Based on this rule, we count the number of people who voted for each project, and the project that earned the most approval will be funded. The outcome will be buying new books for the library regardless of the voter's vote in this case.

• CASE 2:

Sydney City Council has allocated \$1.1 million to enhance the city, considering three potential projects:

- (1) Creating a new park with a cost of \$1 million.
- (2) Establishing an art center with a cost of \$0.9 million.
- (3) Constructing a new shopping center with a cost of \$1.1 million.

With the available budget, the city can only fund one out of these three projects. To make this decision, the City Council has selected three representatives, and your role is to be the third representative, providing your vote.

We considered the first and second representatives are Alice and Bob. They both think the city needs a new park for children, so they

approved the first project and disapproved others. We are using the greedy rule as a method of voting. Based on this rule, we count the number of people who voted for each project, and the project that earned the most approval will be funded. So, in this scenario, the park will be funded regardless of the collected user's vote.

We explained to our participants that they have the option to either approve or disapprove of projects based on their individual interests. There is no restriction on the number of projects that can be approved, but participants must approve at least one project. We then explained that we had already collected votes from other participants and asked them to indicate their approval or disapproval. It was mentioned that all representative votes had been processed by the algorithm. Following their responses, the system's decision will be provided, along with explanations regarding this decision. Subsequently, a series of questions will be presented to assess the impact of these explanations.

After the users' votes were collected and the outcome of the algorithm was shown to them, we asked some questions regarding the transparency of the outcome and the level of the users' trust in the system. In the next stage, we presented explanations and reasoning based on axioms to justify the outcome of the algorithm. Following the presentation of these explanations, we asked further questions to assess the effect and performance of the explanations on the users' understanding of the system.

5 RESULTS

We conducted our pilot study with 26 users (male = 16, female = 10). 62% participants aged between 25- 34 and 23% aged between 35-44. Of the participants, 50% were from India, 42.3% from the United States, and the rest from various countries such as Italy and Sri Lanka. Overall, 42% participants described their computer science literacy as advanced and 31% expert.

Most of the participants (72.4%) found the explanations either extremely helpful or very helpful in understanding the system's decision. Additionally, 63.3% of the participants preferred general explanations over counterfactual ones. About 30% of participants reported that the explanations had a positive impact on their perception of the system's fairness. None of the participants reported a negative impact from the explanations in that matter. However, we could not draw any meaningful conclusions about how the explanations affected users' trust in the system. In some cases (20%), providing the explanations decreased the level of the trust in users. 27% of participants reported an increase in their level of trust in the system after receiving an explanation, 23% reported a decrease in their level of trust, and 50% expressed that their level of trust did not change. In Fig. 1, we present how system decision transparency, fairness, and user trust in system decisions change before and after receiving explanations for both presented PB scenarios in our survey, shown as case 1 and case 2. We provided users five options to choose from for each criterion of transparency, trust, and fairness. As shown, user understanding of the system (system transparency) and perceived fairness improved in both cases after receiving explanations.

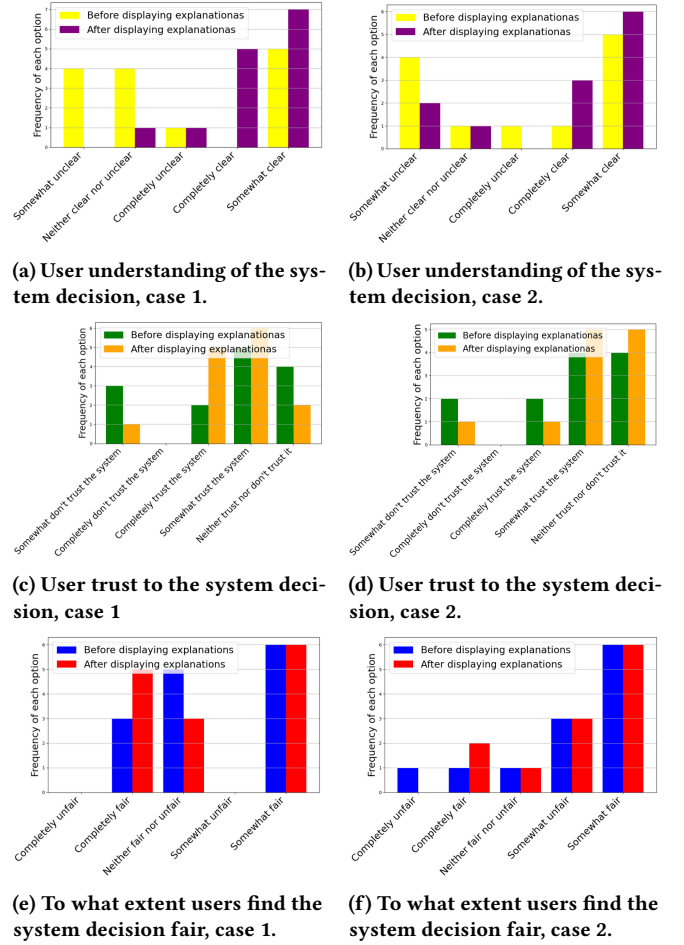


Figure 1: Comparison of user's report regarding the transparency of the system's decision (first row), their trust to the system (the second row), and the fairness (the third row) before and after receiving explanations.

6 CONCLUSION AND FUTURE WORKS

In this study, we focused on providing explanations for AI algorithm outcomes in PB settings. By using axioms (desired properties) as linear constraints, we identified all possible outcomes that meet these criteria, ruling out any that do not. We then tested our approach with real-world users to see if the explanations would enhance transparency, increase user trust, and improve perceived fairness. Our results show that users generally found the explanations helpful for understanding the system, improving transparency and fairness. However, we could not draw a definitive link between trust in the system's decisions and the explanations.

Next, we will conduct an extensive user study with a substantial number of participants to verify our findings and gain deeper insights into the explainable PB setting. Additionally, we will explore the reasons behind instances where providing explanations reduced trust and how to mitigate this issue.

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